A Road-Detail Preserving Framework for Urban Road Extraction From VHR Remote Sensing Imagery

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***Abstract*— Automatic road extraction has gained significant attention in urban navigation, sustainable transport, and disaster response. Conventional convolutional neural networks (CNNs) operate within the local receptive field, limiting their capacity to represent potential global relations between roads and surroundings. In addition, the edge is important topological information for road targets. Several works focus on predicting precise boundaries to enhance road extraction. However, over fit edges and the course integration between features of different network layers may lead to loss of local details and incorrect road segmentation results. Therefore, the Road-detail Preserving Mapper (RoadDP-Mapper) framework is proposed. First, RoadDP-Mapper employs a hierarchical transformer as the encoder to enable local-to-global reasoning. The asymmetric upsampling layers (APLs) are introduced to enhance the model's capability to perceive and reconstruct critical road detail information. Second, a road edge-constrained branch with a detail preservation module (DPM) is devised to amplify the distinction between roads and backgrounds by extracting and preserving explicit class boundary details. The proposed joint loss inspires the transformer to capture the contextual spatial relationships while preserving the fine-grained features of the road. We evaluated our framework on the DeepGlobe dataset and self-annotated images from ten representative cities in China. The proposed framework has demonstrated its effectiveness by significantly reducing both missed detections and false alarms in road extraction. Furthermore, spatial transfer experiments have confirmed the generalizability of RoadDP-Mapper for large-scale road mapping.**

***Index Terms*— Detail preservation, edge constrained, remote sensing, road extraction, transformer.**

I. Introduction

VERY-HIGH-RESOLUTION (VHR) remote sensing imagery, characterized by rich spectral and spatial information, significantly enhances the capabilities of ground object identification and mapping. However, it also introduces challenges in data processing and application [1], [2]. From autonomous driving to urban planning, real-time and accurate road maps play a significant role in various applications [3], [4], [5]. Automatically extracting road information from VHR remote sensing imagery is a prerequisite for large-scale road network extraction [6], [7].

Traditional road extraction methods utilize techniques such as mathematical morphology [8], active contour [9], and artificial neural networks [10]. Due to the complexity and diversity of road feature types in VHR remote sensing imagery, traditional methods can hardly further improve road extraction performance [11], [12]. In recent years, numerous deep learning methods have emerged within the domain of road extraction [13], [14], [15]. Deep learning-based road extraction methods can be classified into patch-based convolutional neural networks (CNNs), fully convolutional networks (FCNs), graph-based neural networks, and generative adversarial networks (GANs) [2], [16]. Among these methods, FCNs with encoder–decoder architectures demonstrate notable adaptability, e.g., U-Net [17] and D-LinkNet [18], and find extensive application [6], [19], [20]. Furthermore, many scholars have introduced attention mechanisms to model the spatial contextual dependencies of roads [21], [22], [23].

Despite the powerful performance, most previous deep neural networks require large amounts of data for training [24], [25], [26], [27]. Therefore, they could be misled to make incorrect predictions when the ground truth contains inaccuracies, such as label scarcity, omission, and noise [28]. In order to mitigate this problem, prior knowledge of road networks is introduced into the road extraction process. For example, roads have well-defined edges, continuous centerlines, and consistent widths within the same scale. Some works [29], [30], [31] fuse prior information to improve road extraction performance. For edge priors, researchers incorporated them to enhance the robustness of neural networks against influences such as occlusion and shading [32], [33]. Liu et al. [34] employed a cascaded deep CNN, where road surface features were used as initial features for the road edge and centerline. Lu et al. [35] developed a CasMT framework to implement road surface segmentation, centerline extraction, and edge detection. Zhou et al. [36] improved the road boundaries by feeding the coarse output into the fine map predicting module. Previous techniques often train a cascade network with binary

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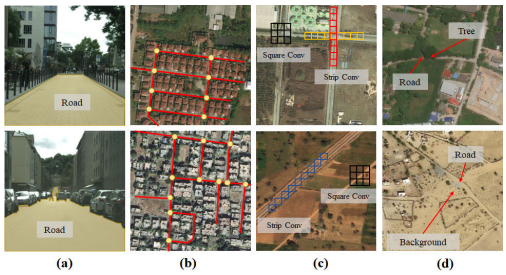


Fig. 1. Examples of road characteristics. (a) In street-view images, road pixels constitute a larger proportion compared to their presence in remote sensing imagery. (b) Road network topology is complex, and it is essential to consider the dependencies between road & road, and road & background. (c) Strip convolution is more suitable than square convolution for extracting narrow linear roads. (d) Road is obscured by trees, and the road is almost invisible among the bare ground. Suitable prior information needs to be incorporated for this problem.

cross-entropy (BCE) loss and output the refined edges of roads. Alternatively, they may directly incorporate low-level features containing boundary information into high-level features. All these works focus on embedding the edge information into high-level features and adapting the precise edge details to revise road segmentation boundaries. Nevertheless, overfit edges will introduce noise and may lead to incorrect road segmentation results.

Methods based on deep learning have made significant explorations in the field of road extraction. However, the task associated with remote sensing imagery presents distinctive characteristics that warrant consideration [37]. Fig. 1 illustrates some remaining problems with road extraction.

1) Compared to natural scenes, roads in remote sensing images exhibit two characteristics. First, as illustrated in Fig. 1(a) and (b), remote sensing images often contain long-span roads that are continuously distributed throughout the image. Road entities may be occluded by trees and buildings, resulting in extraction discontinuity. The dependency between roads and roads, as well as roads and backgrounds, needs to be globally considered. Therefore, long-range dependency is crucial for road extraction. Some CNN-based methods expand the receptive field by stacking the convolution layers and enlarging the convolution kernel to gain a wider range of contextual information [31], [34], [35], [36]. However, convolutions usually operate in the local area, and it is not straightforward to quickly perceive the global dependency information. With its strong ability to model long-range dependency, the Transformer is applied to road extraction tasks [70], [71], [72].
   Second, in remote sensing images, narrow roads take up only a small percentage of the whole image, whereas the complex background that is irrelevant to roads takes up a large part. Directly applying Transformer architecture, which allows each pixel at every position to interact with pixels at other positions, has limitations in fully capturing the strip-like features of roads, as depicted in Fig. 1(c). Therefore, how to effectively model the linear and directional road features when capturing global information still remains to be a challenge.
2) In addition to the challenges brought by the morphological characteristics of roads themselves, under real-world conditions, similarities between roads and other background categories may result in missed detection and misclassification. For example, roads with spectral characteristics similar to those of bare land are difficult to distinguish, as shown in Fig. 1(d). Therefore, it is crucial to extract detailed road edge and texture information to enhance the differentiation between roads and other categories. At road edges, local pixel gray-scale values change rapidly, and this high-frequency information helps recognize the boundaries between roads and background areas [29], [67]. Besides edge information, analyzing relationships between neighboring pixels provides local texture information. The flat surface of the road makes its texture more homogeneous and smooth compared to the surroundings, and this texture difference further aids in distinguishing the road from the background. Although detailed, low-level local information provides accurate spatial location, the road extraction capability at a single scale is limited. Therefore, existing road extraction methods have designed multiscale networks to aggregate local and global information. Skip connections [34], [35], [36], [48] are adopted to connect high-resolution feature maps from the encoder to the upsampled feature maps in order to recover local information. However, direct addition or concatenation [35], [56], [58] fails to match the positions between feature maps, leading to information loss and misalignment [36], [38]. Therefore, more effective feature alignment methods are needed to better preserve road details, facilitate the recovery of road edge and texture information, and accurately distinguish roads from complex backgrounds.

From a global perspective, roads possess elongated and directional characteristics, while, at a local scale, roads exhibit high-contrast details, particularly along their edges. These observations motivate us to integrate road priors into an end-to-end network. We propose a Road-detail Preserving Mapper (RoadDP-Mapper) framework. RoadDP-Mapper employs an encoder–decoder structure. A Transformer-based backbone is constructed to model global information of roads with large spans. In the decoding stage, there are two subnetwork branches designed in parallel. A joint loss inspires the transformer to capture the contextual spatial relationships while preserving the fine-grained features of the road. The main contributions of our work are summarized as follows.

1) To effectively utilize the long-span and directional characteristics of roads, we constructed RoadDP-Mapper, a Transformer-based framework. RoadDP-Mapper leverages the transformer's global information extraction capability to model continuously distributed long-span roads. The APLs module is embedded in the decoder, which targets the reconstruction of linear and directional road information. In contrast to conventional methods which use square convolutions, APLs employ four asymmetric transposed convolutions with different directions, mitigating the impact of complex backgrounds.

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2) To effectively preserve road detail information and enhance the accuracy of predicting road edges, a relaxed edge-constrained branch embedded with the detail preservation module (DPM) module is designed. The rough edge generated by the edge detection branch amplifies the contrast between the road and the background. DPM module consists of the road feature selection mechanism (RSM) and the road feature alignment mechanism (RAM). Among them, the RAM is used to learn pixel transformation offsets that reduce spatial misalignment, while the RSM emphasizes rich local details of roads. Thus, the rich edge information of the road is preserved, and the edge information and nonedge information are effectively fused.
3) Extensive experiments on the DeepGlobe and self-annotated datasets demonstrate the effectiveness of the method, achieving the highest accuracy. In addition, the road mapping results in three cities showcase the excellent generalization ability of RoadDP-Mapper, indicating its high potential for large-scale road mapping applications.

II. Related Work

For patch-based CNN models, the typical procedure involves slicing the input image into patches and subsequently feeding these small patches into successive convolutional layers for feature extraction [38], [39], [40]. FCN models perform end-to-end prediction and have become the underlying architecture for most road extraction models [38], [41], [42]. Among them, encoder–decoder architecture has attracted considerable research attention. The encoder is usually a pretrained classification network, such as VGG [43], [44], ResNet [45], [46], [47], [48], DenseNet [49], and other backbones for feature extraction. The feature reinforcement modules, such as dilated convolution [18], [48], [50], atrous spatial pyramid pooling (ASPP) [30], [51], [52], and others [36], [53], [54] for multiscale feature integration. The decoder upsamples the output of the feature reinforcement module while fusing the features of the encoding layer through skip connections and, finally, obtaining a prediction map of the same size as the original image [55].

Multitask and multimodal methods enhance performance by incorporating supplementary supervision. Multitask research addresses multiple tasks that exhibit complementary characteristics jointly. RVMNet [29] enhances the connectivity of the road mask by incorporating supervised information in the form of road nodes. These nodes obtained through a node proposal network are subsequently inserted into the road vector map, aiming to improve the overall connectivity of the road network. Multimodal research aims to effectively leverage remote sensing data across various modalities. I^2HN [56] introduced an intuition-inspired hypergraph network. A hypergraph parser was designed to learn intramodal objectwise relationships, enabling the generation of robust monomodal representations. A hypergraph matcher was developed to enhance cross-modal compatibility during the fusion of multimodal features.

Over the last decade, CNN-based methods have been predominant in road extraction [18], [36], [50], [57]. Other than conventional CNNs, attention mechanism-based CNN have been introduced to model global contextual relationships [58]. ADS-Net [23] and DBRANet [47] analyzed the potential advantages when incorporating channel and spatial attention mechanisms into road extraction and combined the convolutions with attention blocks. GMRNet [59] designed the GC block attention module to enhance the emphasis on global information. Most of them alleviate the problem of road disconnection of CNNs to some extent.

Recently, pure transformers, such as a Vision Transformer and a Swin Transformer, have emerged as potential alternatives to CNN-based methods in the field of computer vision [60], [61]. Jiang et al. [62] developed a deformable transformer to adaptively explore the spatial relevance of feature recognition. CasFormer [63] is designed with cascade-attention blocks, integrating spectral-aware self-attention and spatial-fused cross-attention, allowing for effective fusion in both spatial and spectral domains. LRR-Net [64] integrates the low-rank representation model into the deep neural network as prior knowledge, leveraging its powerful separation capability. Some scholars have utilized transformers to propose remote sensing foundation models [65], [66], which aid in acquiring comprehensive feature representations for various downstream remote sensing interpretation tasks.

III. Methods

The proposed RoadDP-Mapper is designed as a dual-branch encoder–decoder, which is illustrated in Fig. 2. First, the tiny version of Swin Transformer (Swin-T) [67] is utilized as the encoder. In the Swin-T encoder, the patch merging layer is responsible for creating a hierarchical road representation. The Swin Transformer block (STB) performs local-global information interaction. The features output by Swin-T are then fed into two decoders: the road extraction branch to extract the road map and the edge-constrained branch to extract the feature edge map. In the road extraction branch, APLs are used as upsampling layers. It is more suitable for roads with large spans and narrow-width shapes. In the edge-constrained branch, the edge masks are generated by the pretrained richer convolutional feature (RCF) network. The upsampling path receives the low-level features from the Swin-T encoder as its additional input. The upsampled features and the underlying detail features are connected through the DPM, which is divided into two modules: RAM and RSM. RAM is used for feature alignment and learning transformation offsets of pixels, thereby reducing category misclassification in road prediction results; RSM is used for feature selection and emphasizing road edge detail features. Furthermore, the upsampled edge feature maps are concatenated to the road extraction branch as a guide for refining the road probability maps. The same DPM is used for the interaction of edge branches and road branches. Finally, the joint loss function consisting of road segmentation loss and edge loss is designed to optimize the training.

A. Hierarchical Transformer-Based Backbone

The Swin Transformer is a strong baseline for road extraction, which introduces visual priors such as hierarchical, local,

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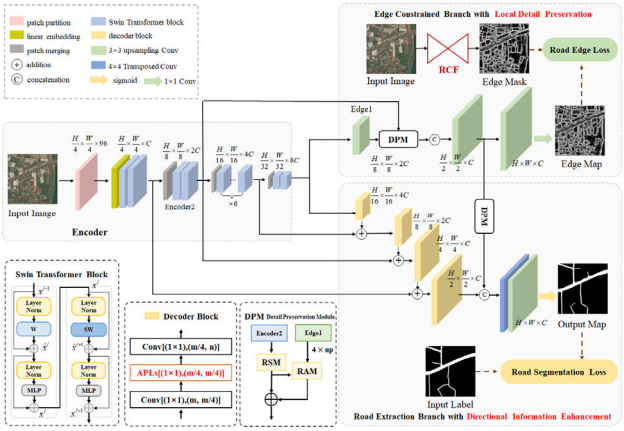


Fig. 2. Overview of RoadDP-Mapper.

and shift-invariant information. The Swin-Tiny pretrained on ImageNet is applied. The input image size is $H \times W$, which is first divided into 4×4 patches. After linear embedding, the size of the feature map is $(H/4) \times (W/4) \times C$. Inspired by the hierarchical structure of CNNs, the Swin Transformer generates pyramid-like feature maps through patch merging. Patch merging downsampled feature maps and feeds them into the next STB. After patch merging, stage-1, stage-2, stage-3, and stage-4 output feature maps of size $(H/4) \times (W/4) \times C$, $(H/8) \times (W/8) \times 2\,C$, $(H/16) \times (W/16) \times 4\,C$, and $(H/32) \times (W/32) \times 8\,C$, respectively. The number of STBs in each of the four stages is 2, 2, 6, and 2.

As shown in Fig. 2, an STB consists of layer norm (LN) layers, a multiheaded attention mechanism module, residual connections, and a two-layer MLP with GELU nonlinearity. W-MSA and SW-MSA modules are cascaded in an STB. Their receptive fields are illustrated in Fig. 3. Compared to the previous Vision Transformer, where every pixel at each position can interact with pixels at other positions, the Swin Transformer adopts the Shift-Window concept. Under this framework, pixels within each window can still interact with one another, while pixels from different windows no longer have direct interactions. By enabling pixel interactions through shifted windows, this approach effectively reduces the amount of computation. The SW module is designed to help capture the rich semantic information and establish relationships between patches. The W module is defined as

$$\hat{x}^{l} = \text{W-MSA}\big(\text{LN}\big(x^{l-1}\big)\big) + x^{l-1} \tag{1}$$

$$x^{l} = \text{MLP}\big(\text{LN}\big(\hat{x}^{l}\big)\big) + \hat{x}^{l} \tag{2}$$

where $\hat{x}^{l}$ and x^{l} represent the output of the W module and the MLP module of the l block, respectively. The SW module can be calculated as

$$\hat{x}^{l+1} = \text{SW-MSA}\big(\text{LN}\big(x^{l}\big)\big) + x^{l} \tag{3}$$

$$x^{l+1} = \text{MLP}\big(\text{LN}\big(\hat{x}^{l+1}\big)\big) + \hat{x}^{l+1} \tag{4}$$

where $\hat{x}^{l+1}$ denotes the output generated by the SW module of the $l+1$ block.

W-MSA and SW-MSA denote window-based multihead self-attention using regular and shifted window partitioning configurations, respectively. Specifically, SW-MSA connects

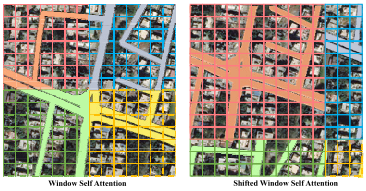


Fig. 3. Illustration of receptive fields in W-MSA and SW-MSA. SW-MSA allows for cross-window information interaction.

some positions between adjacent windows to capture global contextual information more effectively. Through the cascade mode of these modules, the self-attention computation in the new windows extends beyond the boundaries of the previous windows in the layer, thereby establishing long-range dependencies among them.

B. Road Extraction Branch With Directional Information Enhancement

The road extraction branch consists of five decoding stages. The first four decoding stages are made up of the same decoder blocks. The Swin Transformer encoder generates hierarchical feature maps of four scales throughout four stages. To comprehensively integrate multigranularity features, the road extraction branch aggregates the four hierarchical feature maps in the four decoder blocks.

APLs in the decoder blocks are introduced to progressively recover the dimensions. Skip-connection is employed to integrate the corresponding layers. In the fifth decoder layer, the output of the edge-constrained branch and the fourth decoder block are used as input. Then, the 4×4 transposed convolutional layers with stride 2 and the 3×3 convolutions reshape the feature maps to the resolution original resolution. Finally, the road binary map is output after a sigmoid function.

In 3×3 transposed convolution layers, interference from complex backgrounds is often encountered. APLs are more suitable for long, thin, spanning, and small feature targets, such as roads. It consists of four asymmetric transposed convolutions in the vertical, horizontal, positive diagonal, and negative diagonal directions and is represented as (5). Given an input feature F, the output feature $\hat{F}$ of APLs is computed as (6)

$$\text{APLs} = \left[\text{Conv}_{1\times 9}, \text{Conv}_{9\times 1}, \text{Conv}_{\text{diag}}, \text{Conv}_{\text{diag}\sim}\right] \tag{5}$$

$$\hat{F} = \text{Concat}(\text{APLs}(F)). \tag{6}$$

The number of filter parameters in each direction is one-quarter of the number of a 3×3 transpose convolution. As a result, they produce the same number of embeddings. As presented in Fig. 4, the transposed convolutions in all four directions have the same center, and the resulting feature maps are stitched together into one.

C. Edge-Constrained Branch With Local Detail Preservation

During feature representation learning, fine spatial detail is lost in the significantly reduced resolution. To face the challenge, we design an edge constraints branch as an auxiliary classifier to perform edge information embedding. It aims to

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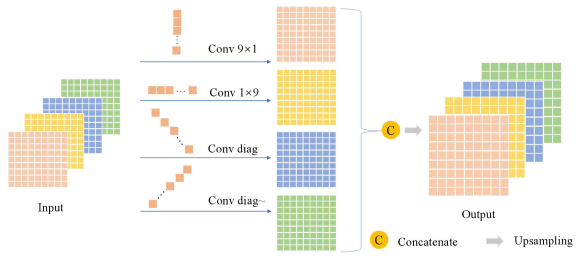


Fig. 4. Illustration of the APL in road extraction branch. Given that the input feature map size is $9 \times 9 \times m$, after APLs, the output feature map size will be $18 \times 18 \times n$.

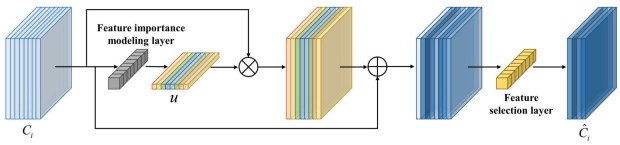


Fig. 5. Road feature selection mechanism.

improve the separability of the road from the background by enhancing the representation with edge aware. To obtain the fine-grained edge prior, the RCFs network [68] is used to predict the edge masks for the training road dataset. In particular, RCF uses the VGG-16 as the backbone, and is pretrained on a mixture of data from the BSD500 and PASCAL VOC Context datasets.

After obtaining the fine-grained latent representation, the output of stage-4 is fed into the edge-constrained branch. The spatial dimension of the feature map is extended via the CARAFE operator [69]. The resolution is increased to $4\times$ of the input, and the dimension remains the same. Then, the feature map is combined with the encoder stage-2 output through DPM.

DPM consists of RSM and RAM. The RSM serves the purpose of accentuating crucial feature maps that encapsulate intricate spatial details to ensure precise allocations, concurrently attenuating superfluous feature maps. As illustrated in Fig. 5, first, the global information of the feature map is extracted through global average pooling and sigmoid function; then, the importance of the feature map is modeled through a 1×1 convolution to obtain the attention map; the attention map is obtained by multiplying the original feature map importance vector; and the addition of the importance vector and the original feature map completes the selection of rich spatial detail features.

For the RAM module, due to the downsampling operation in the encoder stage, there is an unavoidable spatial misalignment between the upsampled feature map of the edge branch and the features of the encoder stage. Hence, feature fusion, whether through elementwise addition or channelwise concatenation, may lead to the loss of road boundaries and texture details, making it difficult to accurately distinguish roads from complex backgrounds. In this module, spatial location information is conveyed through 2-D feature maps, with each offset value representing the shifted distances in 2-D space between individual points in the upsampled feature maps and their corresponding points in the output feature maps

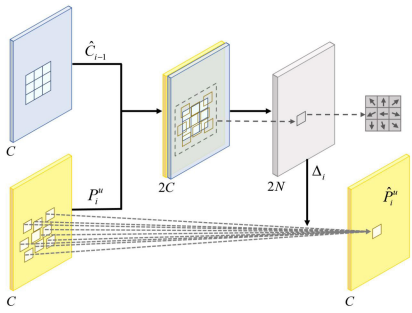


Fig. 6. Road feature alignment mechanism.

of the encoder. As illustrated in Fig. 6, the RAM receives the two feature maps $\hat{C}_{i-1}$ and P_i^u selected by the RSM module to obtain their spatial differences and then learns the offset between the two features through the convolution operation; subsequently, deformable convolution operations are applied to align the two features using learned offsets.

The connected feature map obtained through DPM is again passed to the CARAFE operator. F_{edge} is denoted as the obtained edge embeddings. It is then aggregated to the last decoder layer in the road extraction branch through DPM. Given that the corresponding segmentation embedding is F_{seg}, the output features F_{merge} can be defined as (7). F_{merge} is used to generate road segmentation predictions

$$F_{\text{merge}} = F_{\text{edge}} + F_{\text{seg}}. \tag{7}$$

After $2\times$ bilinear upsampling, the edge embedding F_{edge} is recovered as $H \times W \times C$. After using the 1×1 convolution layer, the predicted edge maps are output.

D. Joint Loss Function

The joint loss function consists of two components, segmentation loss and edge loss, which are defined as

$$L = L_{\text{seg}} + \alpha L_{\text{edge}} \tag{8}$$

where α is a tradeoff coefficient. BCE and DenseCRF loss are summed to form the segmentation loss, which is defined as

$$L_{\text{seg}} = L_{\text{BCE}} + \lambda L_{\text{DenseCRF}} \tag{9}$$

where λ is the weight balance parameter. BCE loss is expressed as

$$L_{\text{BCE}} = -\sum_{p\in\Omega} u_p \log S_p + (1-u_p)\log(1-S_p) \tag{10}$$

where Ω is the set consisting of all pixels, the ground truth u_p, and the prediction S_p as sampling on Ω. To enhance the assessment of pixel-to-pixel resemblance within an image, the DenseCRF loss is employed to leverage the similarity in the data structure. It is a very basic quadratic relaxation and can be calculated as

$$L_{\text{DenseCRF}} = S_p' W (1 - S_p) \tag{11}$$

where $S_p \in [0,1]^{|\Omega|}$ is a soft support vector for all points $p \in \Omega$ and the affinity matrix W is calculated by the sum of Gaussian kernels.

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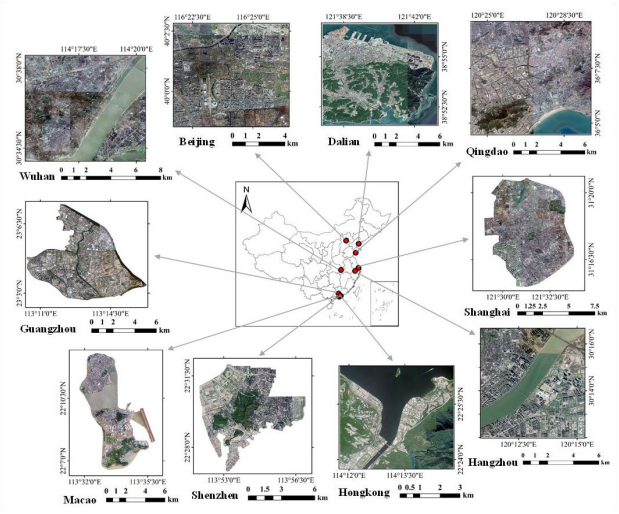


Fig. 7. Overview of the ten urban areas of the self-annotated road dataset.

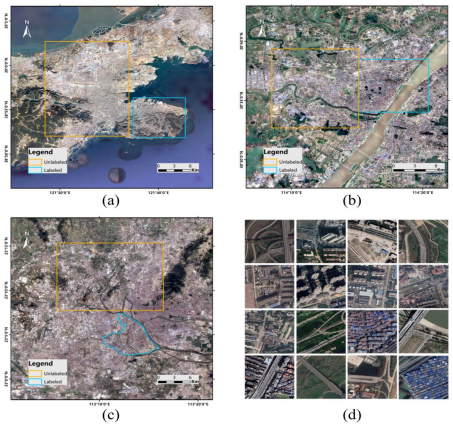


Fig. 8. Visualization of road extraction study area. The blue box indicates the labeled area in the self-annotated dataset. The orange boxes indicate unlabeled areas used for spatial transfer experiments. (a) Dalian study area. (b) Wuhan study area. (c) Guangzhou study area. (d) Examples of different road categories in unlabeled areas.

MSELoss is the loss function L_{edge} for the edge-constrained branch

$$L_{\text{edge}} = \frac{1}{w \times h} \sum_{i=1}^{w} \sum_{j=1}^{h} \left(E_{ij} - B_{ij}\right)^2 \tag{12}$$

where w and h represent the width and height of the image, respectively, E is the edge prior mask, and B is the output edge probability map.

IV. Experimental Results

A. Dataset Descriptions

1) DeepGlobe Road Dataset [70]: This dataset is from the DeepGlobe 2018 CVPR Challenge with pixel-level labels. The original dataset contains 6226 training images, as well as 1243 validation and 1101 testing images, respectively. Each image has a resolution of 1024 × 1024 pixels, with spatial resolution of 50 cm per pixel. To reduce the calculation of the model, we cropped the annotated samples to 512 × 512 sizes. The cropped training set is split into 18 784 training images, 6120 validation images, and 3132 test images. All the experimental results were evaluated on the test dataset.

2) Self-Annotated Road Dataset: This road dataset is a large-scale and high intraclass diversity benchmark for road extraction in Earth vision. It covers roads in major districts of ten Chinese cities, including Beijing, Shanghai, Wuhan, Shenzhen, Hong Kong, Macau, Guangzhou, Hangzhou, Qingdao, and Dalian, which is illustrated in Fig. 7. All VHR remote sensing imageries are collected from Google Earth (Google Inc.). These images have 0.5-m spatial resolution and are 512 × 512 pixels in size. This article uses pixel-level road labels. The numbers of train/validation/test sets are 6015, 962, and 1043, respectively.

3) Study Area for Space Transfer Experiments: The regions of Dalian, Wuhan, and Guangzhou in China, which do not overlap with the annotated regions in the self-annotated dataset, are chosen as a case study for road extraction to test the mapping transfer capacity of the proposed framework. These three cities are located in the northeastern, central, and southern regions of China, respectively, showcasing the country's diverse geographical features. The resolution of the images from all three cities is 50 cm/pixel. The image dimensions are 26 624 × 33 280 pixels for Dalian, 20 992 × 20 992 pixels for Wuhan, and 35 382 × 22 528 pixels for Guangzhou. The specific locations are shown in Fig. 8(a)–(c). There are various types and sizes of roads in the area, as displayed in Fig. 8(d). Road types include motorway, primary, secondary, tertiary, residential, steps, track, and so on. Furthermore, roads in this study area may be obscured by trees and building shadows or confused with surroundings that have similar textures. This better reflects the challenges of large-scale road extraction tasks in practical applications.

B. Experimental Settings and Evaluation Metrics

All models are implemented in Python using PyTorch and trained on four NVIDIA-RTX3090 GPUs. The initial learning rate is set to 0.001 and is reduced by a factor of 5 up to three times when the training loss decreases slowly. The batch size is set to 8, and the Adam optimizer is employed. Our model is trained for 300 epochs on DeepGlobe and 240 epochs on the self-annotated dataset. Following Zhou et al. [18], all training images are augmented, and test time augmentation (TTA) is used.

The precision (Pre), recall (Rec), and overall accuracy (OA) are defined as follows: Pre = TP/(TP + FP), Rec = TP/(TP + FN), and OA = (TP + TN)/(TP + FP + TN + FN). In addition, road IoU, mean IoU (mIoU), and $F1$-score are also quantitative evaluation metrics, where road IoU is employed to focus on how well the road category is extracted and mIoU is the mIoU over all two classes (road and background). These can be computed with

$$\text{IOU} = \frac{\text{TP}}{\text{TP} + \text{FP} + \text{FN}} \tag{13}$$

$$F1\text{-score} = \frac{2 \times (\text{Pre} \times \text{Rec})}{(\text{Pre} + \text{Rec})} \tag{14}$$

where TP, TN, FP, and FN represent true positive, true negative, false positive, and false negative, respectively.

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TABLE I
COMPARISONS WITH OTHER ROAD EXTRACTION METHODS ON DEEPGLOBE

Model Name	Pre	Rec	OA	road IoU	mIoU	F1-score	Flops (GMac)	Params (M)
U-Net [17]	78.29	66.15	95.93	55.90	76.81	71.71	**8.78**	39.5
OCR-HRNet[74]	-	-	-	59.20	78.58	-	53.26	12.08
DeepLabv3+[75]	-	-	-	59.75	78.84	-	176.77	41.23
D-LinkNet [18]	81.28	70.15	96.38	60.39	79.19	75.31	33.58	31.1
ImprovedConnectivity [71]	79.31	76.79	96.53	63.98	81.00	78.03	166.14	29.0
SII-Net [53]	77.70	79.76	96.69	64.90	81.50	78.72	13.74	**3.05**
CasMT [35]†	76.16	83.08	-	65.94	-	79.47	-	-
GAMS-Net [21]†	83.14	80.00	-	68.84	-	81.54	-	-
NAT[76]	-	-	-	65.05	81.62	-	33.42	27.76
OARENet[73]†	81.72	81.95	-	69.25	-	81.84	-	-
GCB-Net [72]	83.52	81.71	97.28	70.37	84.42	82.61	33.99	31.62
Ours	**84.70**	**83.27**	**97.48**	**72.39**	**85.49**	**83.98**	40.23	29.04

†: Results of these architectures are drawn from original papers.

C. Experiments on the DeepGlobe Road Dataset

We compare our RoadDP-Mapper with several representative methods. The original implementation with default parameters from the authors is used, except for SII-Net [53], which requires tuning parameters. For ImprovedConnectivity [71], we select StackModule as the training model. GCB-Net [72] and GAMS-Net [21] leverage attention modules to retain global contextual dependencies from VHR remote sensing. CasMT [35] simultaneously extracts the road centerline, edge, and surface. OARENet [73] employs strip convolution in its network design to enhance the awareness of road occlusions.

The corresponding results on the DeepGlobe road dataset are listed in Table I. The RoadDP-Mapper has the best performance compared with the state-of-the-art. It achieves a high road extraction accuracy of 72.21% road IoU, 85.40% mIoU, and 83.36% $F1$-score. As we can see, the proposed method can provide significantly lower missed detection and false alarm rates in road extraction. To be specific, RoadDP-Mapper outperforms U-Net by +16.49%, OCR-HRNet by +13.19%, DeepLabV3+ by +12.64%, D-LinkNet by +11.82%, ImprovedConnectivity by +8.23%, SII-Net by +7.31%, CasMT by +16.31%, GAMS-Net by +16.31%, NAT by +7.34%, OARENet by +3.14% and GCB-Net by +1.84% on road IoU. This suggests that our model has a stronger feature representation for roads.

Fig. 9 shows the visualization results for all the methods. Compared to other road extraction models, the proposed model has fewer false positive and false negative results. It shows excellent robustness in a variety of complex road scenarios. Specifically, in the first and sixth rows of Fig. 9, when the gray-scale difference between the road and its background is minimal, recognizing such types of roads is challenging for other methods. The edge-constrained branch effectively focuses on the differences between the road and the background, allowing for the extraction of roads with similar gray-scale values on the bare land. In the second and fifth rows, where the road backgrounds are cluttered with distracting elements and the roads exhibit thin, narrow, and irregular shapes, other models experience higher rates of missed detection and false alarms in their extraction results. In contrast, our method benefits from APLs, which help suppress the influence of complex backgrounds, resulting in improved performance. In the third row, for wide trunks, all methods yield good extraction results. However, for narrow roads lacking clear boundaries, our method excels in providing accurate intersection locations and closing small gaps. The fourth row demonstrates RoadDP-Mapper's competence in extracting roads partially obscured by vegetation. In the seventh row, the model successfully captures thin, winding paths. Thanks to the DPM module, which preserves road detail information, RoadDP-Mapper accurately extracts complete roads in this scenario. These observations highlight the robustness and superior road connectivity preservation capabilities of RoadDP-Mapper.

Fig. 10 shows some examples of the difference between prediction and ground truth. Compared with other methods, our method gives fine-grained boundary results on roads. As a whole, RoadDP-Mapper not only extracts more complete roads but also maintains finer edge detail.

D. Experiments on Self-Annotated Road Dataset

Table II presents the quantitative results of the self-annotated road dataset. RoadDP-Mapper surpasses other methods, achieving 69.46% in road IoU and 83.35% in mIoU. In addition, RoadDP-Mapper attained the highest $F1$-score, outperforming U-Net by +28.62%, SII-Net by +5.00%, D-LinkNet by +3.05%, RCFSNet by +1.95%, OARENet by +1.40%, and GCB-Net by +0.94%. These results underscore the effectiveness of employing a Transformer backbone for road extraction compared to CNN-based methods. Table II also displays the number of network parameters (Params) and floating-point operations (FLOPs) for the six compared methods on the self-annotated road dataset. It can be observed that the proposed RoadDP-Mapper improves performance without a significant increase in

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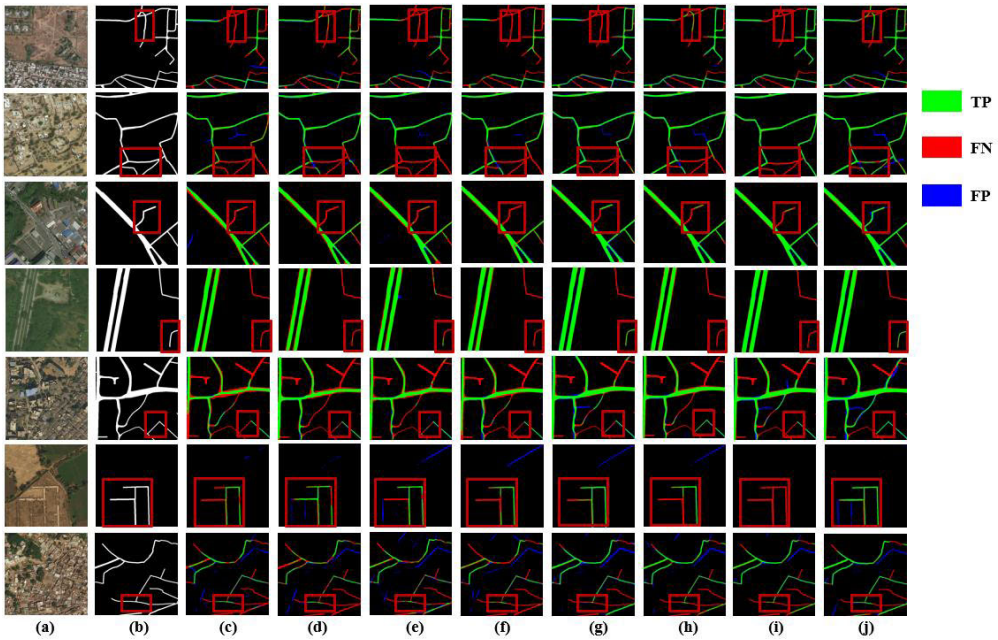


Fig. 9. Road extraction results using the DeepGlobe road dataset. (a) Satellite imagery. (b) Ground truth. (c) U-Net. (d) OCR-HRNet. (e) DeepLabv3+. (f) D-LinkNet. (g) SII-Net. (h) NAT. (i) GCB-Net. (j) Ours.

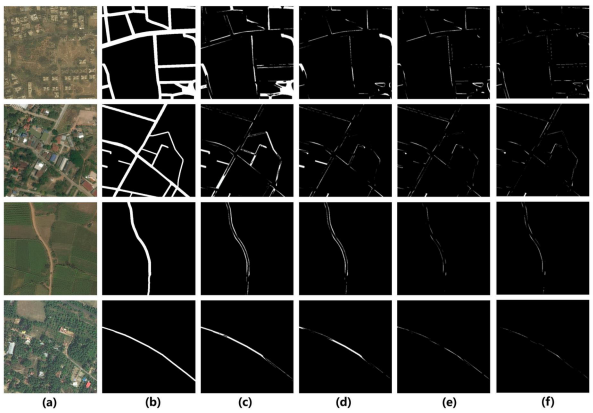


Fig. 10. Illustration of road segmentation results on the boundary. (a) Satellite imagery. (b) Ground truth. (c) D-LinkNet. (d) Ours without edge-constrained branch. (e) GCB-Net. (f) Ours. The fewer white pixels indicate that the gap between the ground truth, and the prediction result is smaller, which means that the road is extracted better at the boundary.

the number of parameters and computational complexity compared with other methods.

Fig. 11 displays several representative visualizations of the self-annotated dataset. As observed in the first, third, and fourth rows of Fig. 11, when the background is complex, GCB-Net, U-Net, D-LinkNet, SII-Net, and Swin-T fail to effectively extract roads. In contrast, the proposed framework successfully extracts more complete roads from building clusters and vegetation despite background clutter, demonstrating the effectiveness of the hierarchical Transformer in capturing long-range dependencies and the impact of the joint loss. When the road is occluded by vegetation or shadow, as shown in the first, third, and fifth rows, RoadDP-Mapper can fill in the missing road segments, maintaining road connectivity better than other methods. In addition, in areas where roads are relatively narrow and complex, as shown in the second and third rows of Fig. 11, the proposed framework exhibits superior extraction performance. This indicates that the proposed DPM can effectively extract slender roads and retain rich detailed information about the roads.

V. Discussion

A. Ablation Study

In RoadDP-Mapper, the APLS module and the DPM module embedded within the road edge branch are proposed. To validate the effectiveness of these modules, various experiments were conducted on the DeepGlobe dataset, as shown in Table III. The baseline model adopts ResNet50 as its backbone architecture within the LinkNet model.

Compared to the experiment in the first row, the addition of the APLS module in the second row resulted in an improvement of 7.49% and 3.91% in road IoU and mIoU, respectively. Consistent with the first innovation, replacing the backbone with Swin-T and adding the APLS module resulted in an increase in road IoU score from 51.69% to 62.47% and mIoU score from 74.60% to 80.25% on the DeepGlobe dataset, as observed from the difference between the first and third rows. This suggests that such a design is capable of guiding the network to focus on critical information while extracting hierarchical road features. Compared to the experiment in the third row, the addition of the edge branch in the fourth row resulted in improvements of 6.98% in road IoU and 3.7% in mIoU. Subsequently, we embedded the DPM module within the edge-constrained branch. As observed from the comparison between the fifth row and the third row, this resulted in improvements in road IoU, mIoU, and $F1$ scores by 9.92%, 5.24%, and 7.08%, thus validating the effectiveness of our second innovation.

Consequently, the final model utilizes Swin-T as the encoder and APLS as the upsampling module for the road extraction

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TABLE II
COMPARISONS WITH OTHER ROAD EXTRACTION METHODS ON THE SELF-ANNOTATED ROAD DATASET

Model Name	Pre	Rec	OA	road IoU	mIoU	F1-score	Flops(GMac)	Params(M)
U-Net [17]	62.45	46.59	89.40	36.39	65.04	53.36	**8.78**	39.5
SIINet [53]	80.28	73.94	93.96	62.57	79.54	76.98	13.74	**3.05**
D-LinkNet [18]	81.71	76.33	94.41	65.19	80.98	78.93	33.58	31.1
RCFSNet [78]	84.54	77.62	94.92	67.97	82.53	80.93	182.3	76.73
OARENet [73]	84.36	77.89	94.93	68.06	82.58	81.00	36.18	33.5
GCBNet [72]	84.44	77.90	94.94	68.12	82.61	81.04	33.99	31.62
ours	**85.30**	**78.90**	**95.16**	**69.46**	**83.35**	**81.98**	40.23	29.04

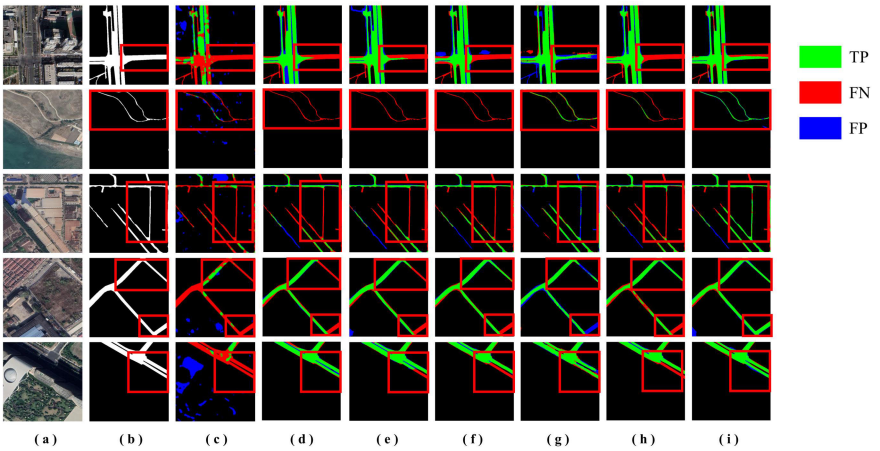


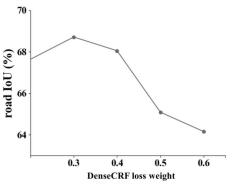
Fig. 11. Road extraction results using the self-annotated road dataset. (a) Satellite imagery. (b) Ground truth. (c) U-Net. (d) SII-Net. (e) D-LinkNet. (f) RCFSNet. (g) OARENet. (h) GCB-Net. (i) Ours.

TABLE III
ABLATION STUDY WITH DIFFERENT COMPONENTS ON DEEPGLOBE

Backbone	APLS	Edge	DPM	road IoU	mIoU	F1-score
ResNet50				51.69	74.60	68.16
	√			59.18	78.51	74.35
Swin-T	√			62.47	80.25	76.90
	√	√		69.45	83.95	81.97
	√	√	√	72.39	85.49	83.98

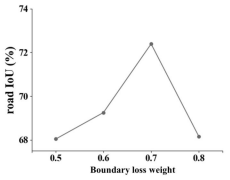
TABLE IV
RESULTS OF ABLATION STUDY WITH THE DIFFERENT LOSS ON THE DEEPGLOBE ROAD DATASET

Edge constrained branch		Road extraction branch		road IoU	mIoU	F1-score
BCE	MSE	BCE	BCE +densecrf			
√		√		64.89	81.51	78.71
	√		√	72.39	85.49	83.98



Weight	Road IoU (%)	mIoU (%)	F1score(%)
0	67.64	83.01	80.70
0.3	**68.71**	**83.56**	**81.45**
0.4	68.05	83.22	80.99
0.5	65.09	81.67	78.85
0.6	64.16	81.18	78.16

(a)



Weight	Road IoU (%)	mIoU (%)	F1score(%)
0.5	68.04	83.21	80.98
0.6	69.25	83.84	81.83
0.7	**72.39**	**85.49**	**83.98**
0.8	68.16	83.28	81.07

(b)

Fig. 12. Effect of different loss weights on the DeepGlobe dataset. All models were trained with epochs of 150. (a) Comparing loss weightings when learning road segmentation. (b) Comparing loss weightings when learning boundary regression and road segmentation.

branch and incorporates edge information. Within the edge branch, the DPM module is employed for detail preservation during intrabranch and interbranch interaction processes. Compared to the baseline model, road IoU, mIoU, and $F1$ scores improved by 20.7%, 10.89%, and 15.82%, respectively. This indicates that the proposed model possesses enhanced feature representation capabilities for road extraction.

B. Analysis of the Joint Loss

To analyze the effect of the joint loss, it is compared by using BCE loss on both branches. From Table IV, we can see that training with joint loss is better than training with BCE loss, resulting in a 7.5% improvement in road IoU and a 3.98% improvement in mIoU. In terms of time efficiency, using BCE loss on both branches will take about 68 h to run 150 training epochs, while using joint loss takes about 54 h in the same

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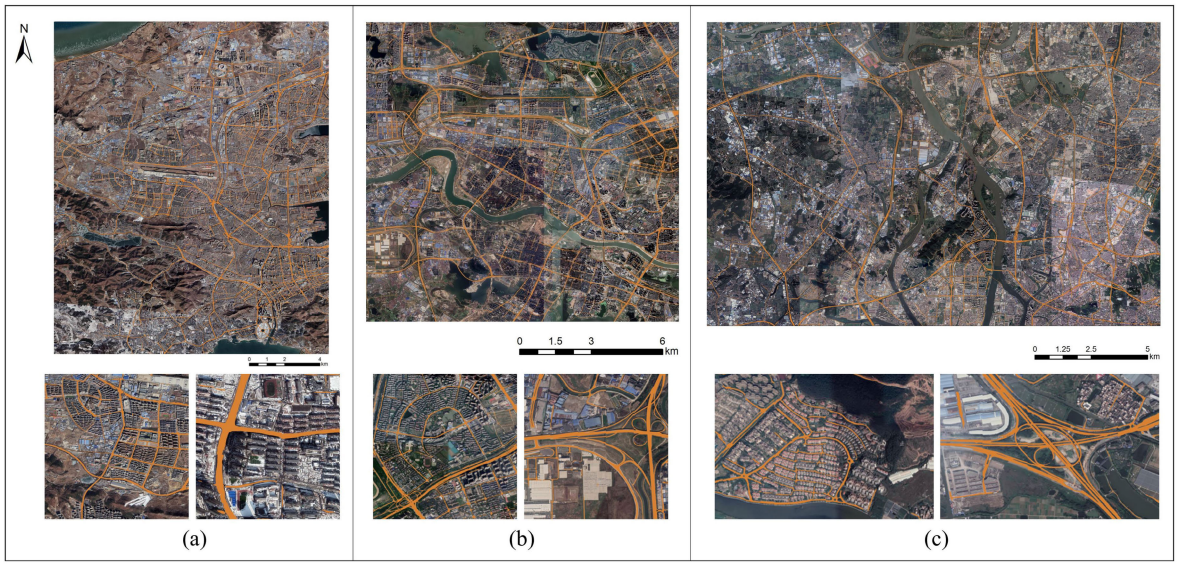


Fig. 13. Visualization of road extraction results. The bottom rows show a zoomed-in view of the road extraction details. (a) Predicted results for the Dalian study area. (b) Predicted results for the Wuhan study area. (c) Predicted results for the Guangzhou study area.

experimental settings. In summary, it is more accurate to use the proposed joint loss for training while being simpler and faster. In Fig. 12, we explore the effect of different λ (9) and α (8) for the DeepGlobe road dataset. We first train a network with only the road extraction branch, as illustrated in Fig. 12(a). BCE loss can easily be misled by inaccurate masks and potentially cause the network to overfit noisy labels. DenseCRF loss refines the boundary prediction during training [77]. Compared with using only BCE loss, combining DenseCRF loss with BCE boosts the accuracy from 67.64% to 68.71% on the DeepGlobe dataset. Overfit edges can introduce noise and lead to unsatisfactory final segmentation results. Motivated by morphological processing and multitask learning, we designed an effective edge-constrained branch. When reporting our best results, we choose $\lambda = 0.7$ and $\alpha = 0.3$.

C. Spatial Transfer Experiments for Road Extraction

Model generalizability is the basis for achieving large-scale road extraction. RoadDP-Mapper trained on the self-annotated dataset is used for spatial transferability test. The study area includes Wuhan, Dalian, and Guangzhou, as illustrated in Fig. 8. The test regions do not overlap with the annotated regions in the self-annotated dataset. As shown in Fig. 13, the proposed method demonstrates good generalization across the three study areas. Despite some instances of isolated roads, missing information, and inaccurate geometry due to shadows and occlusions, the proposed method performs well in narrow road regions and under slight occlusions. This highlights RoadDP-Mapper's ability to model global road information. In addition, RoadDP-Mapper accurately covers road edges, showcasing its capability to preserve road details.

D. Analysis of Failure Cases

Nevertheless, we observe some failed cases of road extraction. As shown in the first and third rows of Fig. 14, when the roads are completely obscured by trees or roads are extremely narrow, the accuracy of our method drops. This may be related to the severe absence of the body and edges of the road. When the road is hidden in a cluttered building or is partially obscured, the RoadDP-Mapper still recognizes them as roads, as shown in the second row. Since the labels do not contain these types of roads, such would be considered false positives. We observe a relatively clear boundary between such roads and the background. This may be one of the reasons why the RoadDP-Mapper can recognize them. Due to the introduction of global context modeling, RoadDP-Mapper prefers to fully connect roads.

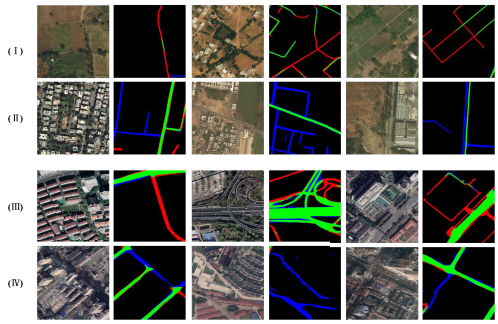


Fig. 14. Visualization of some cases in which our RoadDP-Mapper fails. (I) and (II) Road extraction results on DeepGlobe. (III) and (IV) Road extraction results on the self-annotated road dataset.

VI. Conclusion

In this article, the RoadDP-Mapper framework has been proposed to enhance the model's discrimination of complex road distributions and strengthen the representation of global dependency information in scenes. Our proposed RoadDP-Mapper designs a road extraction branch that enhances local key features, reducing interference from redundant information and facilitating the extraction of narrow and elongated roads. An edge prior-guided branch for road detail recovery and

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preservation is constructed to guide the contraction or expansion of road features, assisting in the extraction of road details. In experiments, the proposed RoadDP-Mapper is optimal compared to other methods on DeepGlobe and self-annotated road datasets, confirming its effectiveness in road extraction tasks. Furthermore, space transfer experiments have demonstrated the potential of the RoadDP-Mapper for large-scale road mapping applications. When significant portions of road bodies and edges are absent, the extraction of complete road segments may still encounter limitations. The integration of graph-based methods could mitigate such issues by focusing on the topological characteristics of roads, thereby enhancing road connectivity. In our future work, more prior knowledge, such as road intersection information, road direction information, or road topological information, will be combined to refine a robust multitask road extraction framework.

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